Supplementary Material

Table S1. Trial-level performance data

Each row corresponds to a single trial. T denotes completion time (s), L denotes master-side trajectory length (m), and the outcome is coded as S (success) or F (failure). Rows are grouped by block and listed in analytical mode order (A–E), not chronological order; the preassigned chronological mode sequence for each operator group is reported in Table S2.

Block ID	Operator ID	Group	Object	Mode	T (s)	L (m)	Outcome
MB01	P01	1	apple	A	19.7647	0.6263	S
MB01	P01	1	apple	B	20.3613	0.5712	F
MB01	P01	1	apple	C	18.2302	0.7079	S
MB01	P01	1	apple	D	20.1232	0.7471	S
MB01	P01	1	apple	E	18.8263	0.7321	S
MB02	P01	1	paper cup	A	20.2387	0.6952	S
MB02	P01	1	paper cup	B	18.8340	0.6637	S
MB02	P01	1	paper cup	C	18.0432	0.5838	S
MB02	P01	1	paper cup	D	19.1196	0.6777	F
MB02	P01	1	paper cup	E	19.9499	0.6712	S
MB03	P01	1	mouse	A	21.9673	0.6332	F
MB03	P01	1	mouse	B	17.3734	0.6484	S
MB03	P01	1	mouse	C	16.4537	0.5699	S
MB03	P01	1	mouse	D	20.7266	0.7682	S
MB03	P01	1	mouse	E	22.2490	0.6481	S
MB04	P01	2	banana	A	19.0863	0.7552	S
MB04	P01	2	banana	B	20.4131	0.8350	S
MB04	P01	2	banana	C	19.5678	0.6649	S
MB04	P01	2	banana	D	20.0671	0.6714	S
MB04	P01	2	banana	E	20.3914	0.8316	S
MB05	P01	2	paper cup	A	25.9611	0.9493	S
MB05	P01	2	paper cup	B	20.1454	0.7550	S
MB05	P01	2	paper cup	C	17.8797	0.6381	S
MB05	P01	2	paper cup	D	20.7920	0.8015	S
MB05	P01	2	paper cup	E	21.0563	0.7940	S
MB06	P01	2	mouse	A	22.1784	0.7902	S
MB06	P01	2	mouse	B	19.3855	0.6696	F
MB06	P01	2	mouse	C	19.6104	0.6812	S
MB06	P01	2	mouse	D	20.6688	0.7218	S
MB06	P01	2	mouse	E	22.0362	0.7554	S
MB07	P01	3	banana	A	21.3677	0.8979	S
MB07	P01	3	banana	B	20.7620	0.7810	S
MB07	P01	3	banana	C	20.6135	0.6701	S
MB07	P01	3	banana	D	21.3607	0.7123	S
MB07	P01	3	banana	E	18.8917	0.7928	S
MB08	P01	3	bottle	A	21.1942	0.7547	S
MB08	P01	3	bottle	B	20.6644	0.6744	S
MB08	P01	3	bottle	C	19.5792	0.6695	S
MB08	P01	3	bottle	D	20.9106	0.7339	F
MB08	P01	3	bottle	E	18.3977	0.6920	S
MB09	P01	3	scissors	A	20.6590	0.5952	S
MB09	P01	3	scissors	B	20.8892	0.9875	F
MB09	P01	3	scissors	C	20.4931	0.7578	S
MB09	P01	3	scissors	D	20.8983	0.5998	S
MB09	P01	3	scissors	E	23.6364	0.7184	F
MB10	P02	4	apple	A	20.9540	0.7275	S
MB10	P02	4	apple	B	22.4177	0.7386	S
MB10	P02	4	apple	C	19.9973	0.8459	S
MB10	P02	4	apple	D	20.3035	0.6780	S
MB10	P02	4	apple	E	22.2594	0.8061	S
MB11	P02	4	bottle	A	21.6166	0.6533	S
MB11	P02	4	bottle	B	22.7903	0.8687	S
MB11	P02	4	bottle	C	18.7488	0.6005	S
MB11	P02	4	bottle	D	20.6251	0.7235	S
MB11	P02	4	bottle	E	18.9052	0.6659	S
MB12	P02	4	scissors	A	22.1109	0.8229	S
MB12	P02	4	scissors	B	20.1043	0.7655	F
MB12	P02	4	scissors	C	17.1904	0.6565	S
MB12	P02	4	scissors	D	24.4447	0.6999	S
MB12	P02	4	scissors	E	20.7281	0.6800	S
MB13	P02	5	apple	A	20.6792	0.7170	S
MB13	P02	5	apple	B	20.8904	0.8292	S
MB13	P02	5	apple	C	18.3496	0.8878	S
MB13	P02	5	apple	D	21.1041	0.7061	S
MB13	P02	5	apple	E	20.5426	0.8040	S
MB14	P02	5	bottle	A	21.1828	0.7639	F

Table S1 (continued)

Block ID	Operator ID	Group	Object	Mode	T (s)	L (m)	Outcome
MB14	P02	5	bottle	B	21.4721	0.6454	S
MB14	P02	5	bottle	C	21.3193	0.7792	S
MB14	P02	5	bottle	D	19.9772	0.6640	S
MB14	P02	5	bottle	E	23.9520	0.6377	F
MB15	P02	5	scissors	A	21.1103	0.6776	F
MB15	P02	5	scissors	B	19.7492	0.7763	F
MB15	P02	5	scissors	C	17.5832	0.7542	S
MB15	P02	5	scissors	D	20.9640	0.6468	S
MB15	P02	5	scissors	E	22.8528	0.5349	S
MB16	P02	6	banana	A	20.8514	0.8434	S
MB16	P02	6	banana	B	20.2425	1.0097	S
MB16	P02	6	banana	C	19.1901	0.9025	S
MB16	P02	6	banana	D	22.9636	0.7686	S
MB16	P02	6	banana	E	22.5321	0.9161	S
MB17	P02	6	paper cup	A	23.1561	0.7838	F
MB17	P02	6	paper cup	B	21.0744	0.7871	S
MB17	P02	6	paper cup	C	19.8447	0.5853	S
MB17	P02	6	paper cup	D	22.3201	0.8671	S
MB17	P02	6	paper cup	E	20.6399	0.7132	S
MB18	P02	6	mouse	A	22.8171	0.6597	S
MB18	P02	6	mouse	B	19.1697	0.8023	S
MB18	P02	6	mouse	C	19.6294	0.8223	S
MB18	P02	6	mouse	D	20.7460	0.8665	S
MB18	P02	6	mouse	E	22.5251	0.6652	S
MB19	P03	7	apple	A	20.4341	0.7572	S
MB19	P03	7	apple	B	21.9394	0.8520	S
MB19	P03	7	apple	C	20.4061	0.6901	S
MB19	P03	7	apple	D	21.7007	0.7223	S
MB19	P03	7	apple	E	22.1380	0.7477	S
MB20	P03	7	paper cup	A	20.9114	0.9774	S
MB20	P03	7	paper cup	B	21.4933	0.9173	F
MB20	P03	7	paper cup	C	18.5434	0.7043	S
MB20	P03	7	paper cup	D	20.8499	0.7190	S
MB20	P03	7	paper cup	E	20.6803	0.7488	F
MB21	P03	7	mouse	A	22.0561	0.6905	S
MB21	P03	7	mouse	B	24.5042	0.8406	S
MB21	P03	7	mouse	C	22.2288	0.7984	F
MB21	P03	7	mouse	D	20.3404	0.8234	S
MB21	P03	7	mouse	E	21.4320	0.7381	S
MB22	P03	8	banana	A	20.5840	0.8097	S
MB22	P03	8	banana	B	21.7273	0.9022	S
MB22	P03	8	banana	C	19.6026	0.7299	S
MB22	P03	8	banana	D	20.2005	0.7632	S
MB22	P03	8	banana	E	21.2047	0.8606	S
MB23	P03	8	paper cup	A	21.5467	0.8037	S
MB23	P03	8	paper cup	B	19.7902	0.7035	S
MB23	P03	8	paper cup	C	19.1190	0.6475	S
MB23	P03	8	paper cup	D	18.8053	0.6170	S
MB23	P03	8	paper cup	E	23.5113	0.6767	S
MB24	P03	8	mouse	A	19.7035	0.8352	S
MB24	P03	8	mouse	B	24.2040	1.0190	S
MB24	P03	8	mouse	C	20.8218	0.7430	S
MB24	P03	8	mouse	D	21.2149	0.6921	S
MB24	P03	8	mouse	E	19.9535	0.7165	S
MB25	P03	9	banana	A	18.4814	0.7055	S
MB25	P03	9	banana	B	21.0957	0.9070	S
MB25	P03	9	banana	C	18.4779	0.8467	S
MB25	P03	9	banana	D	20.7293	0.6453	S
MB25	P03	9	banana	E	20.1558	0.8482	S
MB26	P03	9	bottle	A	24.4515	0.9096	F
MB26	P03	9	bottle	B	23.6510	0.7651	S
MB26	P03	9	bottle	C	18.8794	0.6972	S
MB26	P03	9	bottle	D	21.6030	0.8028	S
MB26	P03	9	bottle	E	19.9488	0.8517	S
MB27	P03	9	scissors	A	23.2623	0.7705	S
MB27	P03	9	scissors	B	22.0770	0.8524	S
MB27	P03	9	scissors	C	20.0995	0.6642	S
MB27	P03	9	scissors	D	21.0219	0.9843	F
MB27	P03	9	scissors	E	19.5680	0.6959	S

Table S2. Preassigned chronological mode sequences

Operator ID	Group	Mode sequence
P01	1	A → D → C → B → E
P01	2	B → E → D → C → A
P01	3	C → A → E → D → B
P02	4	D → B → C → E → A
P02	5	E → C → A → B → D
P02	6	A → D → B → C → E
P03	7	B → E → A → D → C
P03	8	C → B → D → A → E
P03	9	D → A → E → C → B

The preassigned sequence was applied to all blocks within the corresponding operator group.

Table S3. Evidence-boundary overview

Claim	Direct evidence in the present study	Supported interpretation
The complete parameter bundle reduces completion time relative to impedance-only reconfiguration.	The paired C–E mean difference was 1.79 s (bootstrap 95% CI, 1.10–2.51 s); 22 of 27 matched blocks favored mode C.	Supported as a system-level, within-sample result under the tested conditions.
Mode C has a higher success rate.	Observed success was 26/27 in C and 24/27 in E. The paired outcomes were 23 both-success blocks, three C-success/E-failure blocks, and one C-failure/E-success block; the exact two-sided McNemar test gave $p = 0.625$.	Descriptive only; the paired success comparison did not establish a mode effect.
Mode C reduces workload.	Raw NASA-TLX was lower in C in all nine operator × strategy units.	Descriptive within the present sample; no population-level workload inference.
The time benefit results from a shorter path or fewer pauses.	The trajectory-length bootstrap interval crossed zero. Pause-count medians were equal, and no threshold-sensitivity analysis was available.	Not established; these measures are exploratory and do not identify a mechanism.
The haptic and gripper settings each contribute independently, or their combination is synergistic.	Modes C and E changed the haptic-interface and gripper parameter groups together; no factorial channel-ablation design was included.	Not tested.
Configuration was completed before contact in every trial, or the implementation is hard real-time.	Formal trials lacked synchronized trigger–contact timestamps, and hard real-time performance was not evaluated.	Not verified.